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1 Autonomic and somatic organization

COMP

You should be able to: Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the somatic motor pathway and the autonomic pathway side by side, both starting in the CNS and ending at an effector. Count the neurons in each and put the ganglion in its correct place on the autonomic one. Label the transmitter released at every synapse and at both endings. Name the effector tissue at the end of each. One line: what the extra neuron buys the autonomic system.
- ☐ **B** Draw one somatic motor neuron reaching skeletal muscle, and one autonomic pathway branching to cardiac muscle, a smooth muscle, and a gland. Then cut every nerve on your drawing and show what happens to each effector. One line: one of these effectors is completely paralyzed by the cut while the others keep working, say which and why.

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2 Sympathetic and parasympathetic divisions

COMP

You should be able to: Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the CNS as one tall outline from brainstem to sacrum and mark the outflow region for each division on it. From each region, draw a preganglionic and a postganglionic neuron with the ganglion at its true distance from the cord, and make the fiber lengths honestly different between the two divisions. One line: what the ganglion position does to how widely one preganglionic neuron can spread its signal.
- ☐ **B** Draw a heart and a length of gut side by side. Run a sympathetic fiber and a parasympathetic fiber to each, four pathways in total. On each pathway, write the effect with a direction, what happens to heart rate and what happens to gut motility and secretion. One line: name the whole body situation each of your two columns is built for.

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3 COMP

Autonomic neurotransmitters and receptors

You should be able to: Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw four complete pathways from the cord out to an effector, sympathetic to a blood vessel, sympathetic to sweat glands, sympathetic to the adrenal medulla, and parasympathetic to the heart. At every synapse and every ending, write the transmitter and the receptor type. One line: name the two places where your labels broke the pattern, and state what the pattern was.
- ☐ **B** Draw four postsynaptic membranes in a row, one each for nicotinic, muscarinic, alpha, and beta receptors. On each, draw the transmitter that binds it, name one real location in the body, and draw the response it produces, contract, relax, speed up, or secrete. One line: two of your four bind the same transmitter, name them and say what makes their responses different.

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4 COMP

Dual innervation and tonic control

You should be able to: Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw an organ with both divisions on it as a dial with a pointer, sympathetic pushing one way and parasympathetic the other. Set the pointer at rest and mark which division is pulling harder there. Beside it, draw a blood vessel, which gets only one division, and show how its diameter is adjusted with a single input. One line: define tone using both of your drawings.
- ☐ **B** Draw a heart with both divisions attached and write its resting rate on it. Then draw it twice more, once with the parasympathetic supply cut and once with the sympathetic supply cut, putting a rate on each and an arrow for the direction it moved. The pacemaker's own intrinsic rate is higher than the resting rate. One line: say what that fact tells you about which division dominates at rest.

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5

COMP

Autonomic pharmacology

You should be able to: Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a body outline with four organs on it, heart, airways, blood vessels, and gut, and label the autonomic receptor type that dominates at each. Then draw a non selective beta blocker arriving and mark every organ it lands on with the effect it produces there. One line: name the organ where this drug does something the prescriber did not want, and the receptor responsible for it.
- ☐ **B** Pick a muscarinic antagonist and draw its effect at four places at once, pupil, salivary gland, heart, and gut. In each drawing, show the parasympathetic input being blocked and the change that follows. Then write which of the four you were actually treating and which three are side effects. One line: what would have to change about the drug to keep only the effect you wanted.

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6

COMP

Autonomic function testing

You should be able to: Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a heart rate trace across several breathing cycles, with the breathing cycle drawn underneath on the same time axis. Show heart rate rising and falling against inspiration and expiration, using your own recorded data, and mark the beat to beat variation you measured. One line: name the division being withdrawn and restored fast enough to make that pattern, and say how the speed alone tells you.
- ☐ **B** Draw a time axis running through one challenge you performed, cold pressor or standing up. Plot heart rate and blood pressure on it as two curves, before, during, and after, using your own numbers, and mark the moment of the challenge. One line: read your own curves and say which division took over, naming the receptor doing the work out at the effector.

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7
COMP

Hormone classes

You should be able to: Classify hormones as peptide, steroid, or amine and compare the three by synthesis, storage, and receptor location.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw three production lines side by side, one peptide, one steroid, one amine. On each, draw the raw material it starts from, whether the cell stores it in vesicles or makes it on demand, how it gets out of the cell, and where its receptor sits, on the surface or inside. Keep all three at the same scale. One line: two of the three lines look nearly identical, so name which two and the single step where they diverge.
- ☐ **B** The amine class does not behave as one class. Draw a catecholamine and a thyroid hormone side by side, both built from tyrosine, and follow each one out of its cell, through the blood, and to its receptor. Show the point where they part company. Beside each, write which of the other two classes it now resembles. One line: name the property of the molecule itself that decided which way each one went.

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8
COMP

Hormone transport and half life

You should be able to: Explain how protein binding in plasma affects the free hormone fraction and the half life and predict which hormone class circulates bound.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a length of capillary with a steroid hormone and a peptide hormone both inside it. Draw the steroid mostly stuck to carrier proteins with a small free fraction, and the peptide traveling free. At the tissue, draw an arrow for which molecules can actually leave and act. At the kidney, draw which ones get filtered. One line: only free hormone acts and only free hormone is cleared, so say why binding makes a hormone last longer.
- ☐ **B** Draw plasma concentration against time after one pulse of a bound hormone and one pulse of an unbound hormone on the same axes, and mark the half life on each. Then draw a third case, the bound hormone in a person whose carrier protein level has doubled. Show what happened to total hormone and what happened to free hormone. One line: the total went up but the person feels normal, so name the number the lab should have measured.

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9

COMP

Hormone receptors and target response

You should be able to: Explain why one hormone can produce different responses in different tissues and relate receptor number to target cell sensitivity.

PICK ONE AND DO THAT ONE

☐ **A** Draw one hormone molecule in the middle with arrows out to three different target tissues. Pick a hormone with genuinely different actions in different places, not insulin. In each tissue draw the receptor it meets and the response it gets, and make at least two of the three responses run in opposite directions. One line: the hormone was identical in all three, so name what was different and say where that difference physically lives.

☐ **B** Draw one target cell three times: with a normal number of receptors, after weeks of very high hormone levels, and after weeks of very low hormone levels. Count receptors on the drawing rather than just labeling it. Beside the three cells, draw all three dose response curves on shared axes. One line: one of your cells now needs far more hormone for the same response, so name it and say what the body did to cause that.

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10

COMP

Hormone interactions

You should be able to: Define synergism, permissiveness, and antagonism and identify each in a described hormone pair.

PICK ONE AND DO THAT ONE

☐ **A** Draw the same axes three times, hormone dose on the x and response on the y, and use them to show synergism, permissiveness, and antagonism. On each pair of axes plot hormone A alone, hormone B alone, and the two given together, so the shape of the interaction is visible. Label a real hormone pair on each. One line: in one of the three, hormone B does almost nothing by itself, so name that interaction and say why B still matters.

☐ **B** Pick three hormone pairs, one synergistic, one permissive, one antagonistic, and draw each pair acting on the same target organ, with the organ drawn once per pair. Size the arrows into the organ to the effect each hormone has alone and to the effect the two have together. Then take one hormone out of each pair and draw what is left. One line: in one pair, removing the hormone that does nothing alone wrecked the response, so name that pair.

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11
COMP

Control of hormone release

You should be able to: Classify a hormone stimulus as humoral, neural, or hormonal and trace an example of each.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw three trigger paths on one page, each ending at a gland letting go of its hormone. Make one humoral, driven by an ion or nutrient level in the blood, one neural, driven by a nerve firing onto the gland, and one hormonal, driven by another hormone. Use a real example for each and draw the actual stimulus, not the word for it. One line: name which of your three acts fastest and say what makes it fast.
- ☐ **B** Draw plasma calcium falling below its set point and trace the whole humoral loop it starts, gland, hormone, target tissues, and calcium returning, with a feedback arrow closing the loop. Beside it, draw the neural path to the adrenal medulla, from spinal cord to hormone in the blood, and note that this one has no releasing hormone anywhere in it. One line: say what the adrenal medulla is in tissue terms that lets a nerve release a hormone directly.

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12
COMP

Hypothalamic pituitary axes

You should be able to: Trace a three tier axis from hypothalamic releasing hormone through the anterior pituitary trophic hormone to the peripheral gland and place the long and short loop negative feedback.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a three tier axis of your choosing, not the thyroid one, from hypothalamus to anterior pituitary to peripheral gland to final hormone. Draw the portal vessels between the first two tiers and systemic blood after that, because they are not the same route. Then draw the long loop feedback arrow and the short loop feedback arrow and land each one where it belongs. One line: say what the portal system buys the body that dumping the releasing hormone into the general circulation would not.
- ☐ **B** Draw one axis with its feedback loops and all three hormone levels marked normal. Then draw the same axis in a person who has taken a large daily dose of the final hormone from outside for six months. Show what happened at each tier and draw the peripheral gland at the size it ends up. One line: say what happens on the day that outside hormone is stopped suddenly, and why.

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13 Primary and secondary endocrine disorders

COMP

You should be able to: Use hormone levels at two tiers of an axis to classify a disorder as primary or secondary and as hypersecretion or hyposecretion.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a two tier axis, the pituitary trophic hormone above and the peripheral gland hormone below, and draw it four times: primary hyposecretion, secondary hyposecretion, primary hypersecretion, secondary hypersecretion. In each, draw both levels as bars, high, normal, or low, and mark where the break sits. One line: state the rule all four of your drawings obey, in terms of whether the two bars move the same way or opposite ways.
- ☐ **B** Your patient has low thyroid hormone. Draw the axis twice, once with the fault in the thyroid and once with the fault in the pituitary, and draw the TSH level as a bar in each. Then draw what happens to each patient's TSH once you give them thyroid hormone by mouth. One line: one lab value separated these two patients before you treated either of them, so name it and give its direction in each case.

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14 Posterior pituitary hormones

COMP

You should be able to: Explain why the posterior pituitary is neural tissue and state the stimulus and target action of antidiuretic hormone and oxytocin.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the hypothalamus and both pituitary lobes in one sagittal section. For the posterior lobe, draw the neuron cell bodies up in the hypothalamus with their axons running down into the lobe and ending on capillaries. For the anterior lobe, draw the portal vessels instead. Shade the neural tissue and the glandular tissue differently. One line: say why cutting the stalk does different things to the two lobes' hormones.
- ☐ **B** Draw a person who has been out in the desert all day. Draw plasma osmolarity climbing above the normal 290 mOsm per liter, the osmoreceptor detecting it, the hypothalamic neuron firing, ADH leaving the posterior lobe, and water channels going into the collecting duct membrane. Draw the urine that results. Beside it, draw the oxytocin neuron and both of its targets. One line: both hormones leave the same lobe, so say what decides which one comes out.

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15

COMP

Hormone assay

You should be able to: Run or simulate an immunoassay for a hormone and use the standard curve to determine an unknown concentration.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the standard curve you would build, known hormone concentration on the x axis and measured signal on the y, with your standards as points and a line through them. Then put an unknown sample's signal on the y axis and draw the construction lines that read its concentration off the x. Add a second unknown that lands above your highest standard. One line: say what you do with that second sample and why you cannot simply extend the line.
- ☐ **B** Draw the assay well in cross section at four stages: antibody stuck to the well, hormone captured, detection antibody added, color developed. Draw what a well holding a lot of hormone looks like beside an empty one. Then draw a well where a similar but wrong molecule got captured instead. One line: that last well still gives you a number, so say what the number is actually measuring and what it is not.

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16

COMP

Hypothalamic pituitary gonadal axis

You should be able to: Trace the gonadotropin releasing hormone axis to the gonads and place the feedback loops for both sexes.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the axis from GnRH pulses down through LH and FSH, then split it into two branches at the gonad, male and female. On the male branch, run LH to the Leydig cell and FSH to the Sertoli cell and draw what each produces. On the female branch, run LH to the theca cell and FSH to the granulosa cell and do the same. Draw the negative feedback arrows on both. One line: name the hormone that feeds back on FSH alone in both sexes.
- ☐ **B** Draw the female axis twice, once in the early follicular phase where moderate estradiol feeds back negatively, and once just before ovulation where high estradiol feeds back positively. Use the same diagram both times and change only the sign on the feedback arrow. Show what LH is doing in each. One line: the same hormone on the same axis flipped the sign, so say what was different about the estradiol signal.

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17

COMP

Male reproductive physiology

You should be able to: Explain the control of spermatogenesis by follicle stimulating hormone, luteinizing hormone, and testosterone and state the roles of the Sertoli and Leydig cells.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a cross section through one seminiferous tubule with a Leydig cell outside it in the interstitium. Bring FSH to the Sertoli cell and LH to the Leydig cell, and draw what each cell makes. Draw the tight junctions between Sertoli cells and the germ cells maturing from the outer wall inward toward the lumen. One line: name what those tight junctions keep apart and say why that separation matters.
- ☐ **B** Draw the axis from GnRH down to the testis, then draw both negative feedback arrows coming back, one carrying testosterone and one carrying inhibin, and land each on the tier it actually inhibits. Then draw the same axis in a man taking testosterone from outside. Show what happened to LH, to FSH, to the Leydig cells, and to sperm production. One line: he added the male hormone and his fertility fell, so name the step that explains it.

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18

COMP

Ovarian cycle

You should be able to: Order the follicular, ovulatory, and luteal phases and explain how the luteinizing hormone surge is triggered by positive feedback.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a 28 day timeline and put the follicle on it as a run of drawings: recruited follicle, dominant follicle, ovulation at about day 14, corpus luteum, corpus albicans. Under each drawing write the hormone that structure is making at that moment. One line: the follicular phase varies in length between women and the luteal phase does not, so name the structure whose lifespan is fixed.
- ☐ **B** Draw the positive feedback loop that fires the LH surge as a closed circle: the dominant follicle making estradiol, estradiol running high and staying high for about two days, the hypothalamus and pituitary flipping from inhibited to driven, LH climbing steeply, and that LH driving the follicle harder still. Mark on the circle where ovulation breaks it, about 36 hours after the surge begins. One line: positive feedback cannot stop itself, so name what ends this loop.

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19
COMP

Uterine cycle

You should be able to: Correlate the menstrual, proliferative, and secretory phases of the uterine cycle with the estrogen and progesterone levels that drive them.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the uterine lining in cross section at four points across 28 days: mid menstrual, mid proliferative, just after ovulation, and mid secretory. Draw the real thickness, the glands straight or coiled, and the spiral arteries. Underneath, run estrogen and progesterone curves along the same time axis so each drawing sits under its hormone level. One line: name which hormone built the lining and which one finished it.
- ☐ **B** Draw the endometrium at the end of the secretory phase, then draw the next three days as the corpus luteum dies. Show progesterone falling, the spiral arteries clamping down, and mark the layer that is shed and the layer that stays behind. Label both layers. One line: this lining is lost because a hormone went away, not because a hormone arrived, so say what that means for someone whose corpus luteum gets rescued.

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20
COMP

Pregnancy and placental hormones

You should be able to: Explain how human chorionic gonadotropin rescues the corpus luteum and state the roles of placental estrogen, progesterone, and relaxin.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a cycle ending the usual way with the corpus luteum dying, then draw a second version where a blastocyst has implanted. In the second, draw the trophoblast making hCG, the hCG reaching the corpus luteum, and the corpus luteum surviving. Run progesterone along the bottom of both. One line: hCG works by binding a receptor that was already sitting on the corpus luteum for a different hormone, so name that hormone.
- ☐ **B** Draw one time axis from conception to birth and plot hCG, progesterone, and estrogen across the whole of it. Mark the handover in the first trimester where the placenta takes over progesterone production from the corpus luteum. Beside the axis, draw what progesterone, estrogen, and relaxin are each doing to the uterus, the cervix, and the pelvis. One line: say what would happen if that handover failed and the corpus luteum died on its usual schedule.

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21

COMP

Parturition and lactation

You should be able to: Explain labor as a positive feedback loop driven by oxytocin and compare the hormonal control of milk production with milk ejection.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw labor as a closed positive feedback circle: the fetal head pressing on the cervix, stretch receptors firing, the signal reaching the hypothalamus, oxytocin released from the posterior pituitary, the myometrium contracting, and the head driven harder into the cervix. Draw the loop three times round with the arrows getting thicker each turn. One line: name the event that breaks this loop and say why nothing short of it could.
- ☐ **B** Draw the breast in cross section with the alveoli and the ducts, and run two separate control lines into it: prolactin from the anterior pituitary to the alveolar cells making the milk, and oxytocin from the posterior pituitary to the myoepithelial cells squeezing it out. Show suckling setting off both through the same afferent nerve. One line: a mother can be making plenty of milk and still not deliver it, so name which of the two failed and what stress has to do with it.

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22

COMP

Hormone cycle graph interpretation

You should be able to: Read a graph of gonadotropin and ovarian hormone levels across a cycle and identify the day of ovulation and the event driving each peak.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one set of axes covering 28 days and plot four curves on it, LH, FSH, estradiol, and progesterone, labeled along their length so they can be told apart. Drop a vertical line on the day of ovulation. Beside each peak, write the structure that was making that hormone at that moment. One line: name the peak you used to place ovulation, and say whether ovulation comes before or after it.
- ☐ **B** Draw the four curves for a normal cycle, then draw a second set of axes for a cycle in which ovulation never happened. Change only what has to change and leave the rest alone. Show which curve has lost its peak and which one never rises at all. One line: name the single hormone you would measure in the second half of a cycle to prove ovulation happened, and say what it should be doing.

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